

DIALING CODE - COLLECTIONS

Problem Statement

You are required to design a **C# utility class** that manages **international telephone dialing codes** using a dictionary-based approach. The system should allow adding, updating, retrieving, validating, and analyzing country dialing codes efficiently.

Students must **implement each task step-by-step**, ensuring correct usage of **Dictionary, methods**, and **return values**.

Overall Design Specifications

Specification Type	Details
Namespace Name	DialingCodesApp
Class Name	DialingCodes
Class Type	public static class
Data Structure Used	Dictionary<int, string>
Data Members	 No instance data members (all methods are static)
Access Modifiers	All methods: public static

Task 1: Create an Empty Dialing Code Dictionary

Item	Specification
Method Name	GetEmptyDictionary
Return Type	Dictionary<int, string>

Input Parameters	None
Purpose	Creates and returns an empty dictionary to store country codes and country names

Expected Outcome

An empty dictionary with **no key–value pairs**.

Input–Output Test Case

Input	Output
No input	<code>{ }</code> (empty dictionary)

Task 2: Create a Dictionary with Predefined Country Codes

Item	Specification
Method Name	<code>GetExistingDictionary</code>
Return Type	<code>Dictionary<int, string></code>
Input Parameters	None
Purpose	Returns a dictionary pre-filled with known country dialing codes

Expected Outcome

Dictionary contains predefined country codes.

Input–Output Test Case

Input	Output
No input	<code>{ 1: "United States of America", 55: "Brazil", 91: "India" }</code>

Task 3: Add a Country to an Empty Dictionary

Item	Specification
Method Name	<code>AddCountryToEmptyDictionary</code>
Return Type	<code>Dictionary<int, string></code>
Input Parameters	<code>int countryCode, string countryName</code>
Purpose	Creates a new dictionary and adds a single country entry

Expected Outcome

Dictionary contains exactly **one entry**.

Input–Output Test Case

Input	Output
<code>countryCode = 81, countryName = "Japan"</code>	<code>{ 81: "Japan" }</code>

Task 4: Add a Country to an Existing Dictionary

Item	Specification
Method Name	<code>AddCountryToExistingDictionary</code>
Return Type	<code>Dictionary<int, string></code>
Input Parameters	<code>Dictionary<int, string> existingDictionary, int countryCode, string countryName</code>
Purpose	Adds or updates a country entry in an existing dictionary

Expected Outcome

Dictionary includes the newly added country.

Input–Output Test Case

Input	Output
<code>{91:"India"}, 44, "United Kingdom"</code>	<code>{91:"India", 44:"United Kingdom"}</code>

Task 5: Retrieve a Country Name Using Country Code

Item	Specification
Method Name	<code>GetCountryNameFromDictionary</code>
Return Type	<code>string</code>
Input Parameters	<code>Dictionary<int, string> existingDictionary, int countryCode</code>
Purpose	Retrieves the country name for a given country code

Expected Outcome

Returns country name if found, otherwise returns **empty string**.

Input–Output Test Case

Input	Output
<code>{91:"India"}, 91</code>	<code>"India"</code>
<code>{91:"India"}, 99</code>	<code>""</code>

Task 6: Check if a Country Code Exists

Item	Specification
Method Name	<code>CheckCodeExists</code>
Return Type	<code>bool</code>

Input Parameters	<code>Dictionary<int, string> existingDictionary, int countryCode</code>
Purpose	Checks whether a given country code exists in the dictionary

Expected Outcome

Returns `true` if present, `false` otherwise.

Input–Output Test Case

Input	Output
<code>{1:"USA"}, 1</code>	<code>true</code>
<code>{1:"USA"}, 44</code>	<code>false</code>

Task 7: Update an Existing Country Name

Item	Specification
Method Name	<code>UpdateDictionary</code>
Return Type	<code>Dictionary<int, string></code>
Input Parameters	<code>Dictionary<int, string> existingDictionary, int countryCode, string countryName</code>
Purpose	Updates the country name only if the code already exists

Expected Outcome

Country name is updated; dictionary size remains unchanged.

Input–Output Test Case

Input	Output
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<code>{91:"India"}, 91, "Republic of India"</code>	<code>{91:"Republic of India"}</code>
<code>{91:"India"}, 44, "UK"</code>	<code>{91:"India"}</code> (unchanged)

Task 8: Remove a Country from Dictionary

Item	Specification
Method Name	<code>RemoveCountryFromDictionary</code>
Return Type	<code>Dictionary<int, string></code>
Input Parameters	<code>Dictionary<int, string> existingDictionary, int countryCode</code>
Purpose	Removes a country entry if the country code exists

Expected Outcome

Dictionary no longer contains the removed country.

Input–Output Test Case

Input	Output
<code>{1:"USA", 91:"India"}, 1</code>	<code>{91:"India"}</code>
<code>{91:"India"}, 44</code>	<code>{91:"India"}</code> (unchanged)

Task 9: Find the Longest Country Name

Item	Specification
Method Name	<code>FindLongestCountryName</code>
Return Type	<code>string</code>

Input Parameters	<code>Dictionary<int, string> existingDictionary</code>
Purpose	Identifies and returns the country name with the maximum length

Expected Outcome

Returns the longest country name or empty string if dictionary is empty.

Input–Output Test Case

Input	Output
<code>{1:"USA", 91:"India", 55:"United States of America"}</code>	<code>"United States of America"</code>
<code>{ }</code>	<code>" "</code>
